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In [3]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# Seaborn is a library built on top of Matplotlib that makes creating beautiful plots easier.
# Load the dataset directly from a URL
df =
pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')
print("--- First 5 Rows ---")
print(df.head())
print("--- Basic Info ---")
# .info() is a great first command. It tells us the column names, how many non-null values are in each column, and their data types. # Notice that 'Age' and 'Cabin' have missing values!
df.info()
# Get summary statistics for numerical columns
print("--- Descriptive Statistics ---")
print(df.describe())
print("--- Key Insights from Statistics ---")
print(f"The average age of a passenger was {df['Age'].mean():.1f} years.")
print(f"The overall survival rate was {df['Survived'].mean():.1%}.")
print(f"Fares ranged from ${df['Fare'].min()} to a whopping ${df['Fare'].max()}.")
sns.set_style('whitegrid')
# Sets a nice visual style for our
plt.figure(figsize=(8, 6))
sns.countplot(x='Survived', data=df)
plt.title('Survival Distribution (0 = Died, 1 = Survived)')
plt.show()
print("Insight: Far more people died than survived. This is an example of an imbalanced dataset, which can sometimes be a challenge for machine learning models.")
plt.figure(figsize=(10, 6))
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title('Survival by Passenger Class')
plt.legend(['Died', 'Survived'])
plt.show()
print("Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compared to 3rd class passengers. Money seems to have made a difference.")
plt.figure(figsize=(10, 6))
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Survival by Gender')
plt.legend(['Died', 'Survived'])
plt.show()
print("Insight: The pattern is undeniable. A much higher proportion of females survived compared to males. This is another very strong predictor.")
# A FacetGrid allows us to create multiple plots side-by-side to compare distributions.
# Here, we create one histogram for passengers who died (col='Survived'=0) and one for those who survived (col='Survived'=1).

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```

g = sns.FacetGrid(df, col='Survived', height=6)
g.map(plt.hist, 'Age', bins=20)
plt.show()
print("Insight: The age distribution for those who did not survive is
centered around the 20-40 age range. For those who survived, there is a
noticeable spike for young children. This supports the 'children' part of
the mantra.")

# --- Experiment 1: Port of Embarkation ---
print("Experiment 1: Port of Embarkation")
plt.figure(figsize=(10, 6))
sns.countplot(x='Embarked', hue='Survived', data=df)
plt.title('Survival Rate by Port of Embarkation')
plt.xlabel('Embarked Port (C = Cherbourg, Q = Queenstown, S = Southampton)')
plt.ylabel('Count')
plt.legend(['Died', 'Survived'])
plt.show()

# --- Experiment 2: Fare vs. Survival ---
print("Experiment 2: Fare vs Survival")
plt.figure(figsize=(10, 6))
sns.boxplot(x='Survived', y='Fare', data=df)
plt.title('Fare Distribution by Survival Outcome')
plt.xlabel('Survived (0 = Died, 1 = Survived)')
plt.ylabel('Fare ($)')
# Setting y-axis limits to better view the interquartile range due to high
outliers ($512+)
plt.ylim(0, 200)
plt.show()

```

--- First 5 Rows ---

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

--- Basic Info ---

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
4	Sex	891 non-null	object
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	object
9	Fare	891 non-null	float64
10	Cabin	204 non-null	object
11	Embarked	889 non-null	object

dtypes: float64(2), int64(5), object(5)

memory usage: 83.7+ KB

--- Descriptive Statistics ---

	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

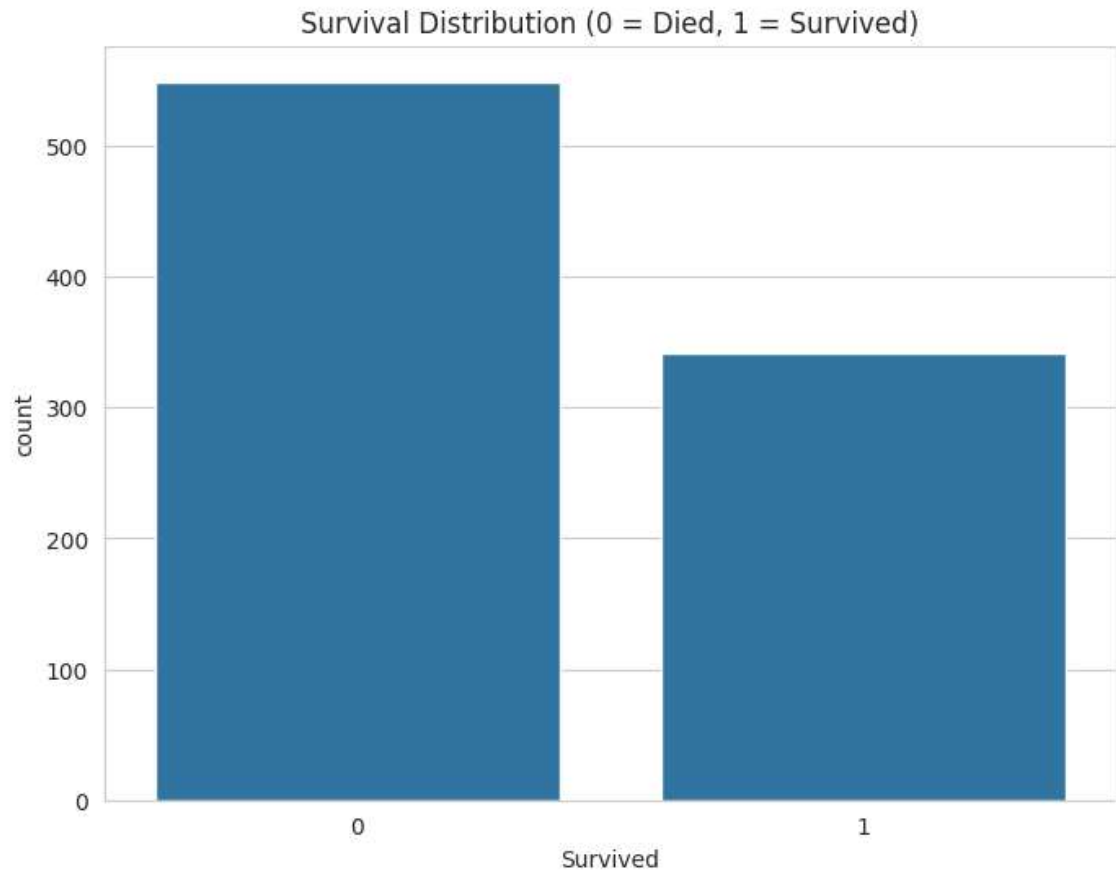
	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

--- Key Insights from Statistics ---

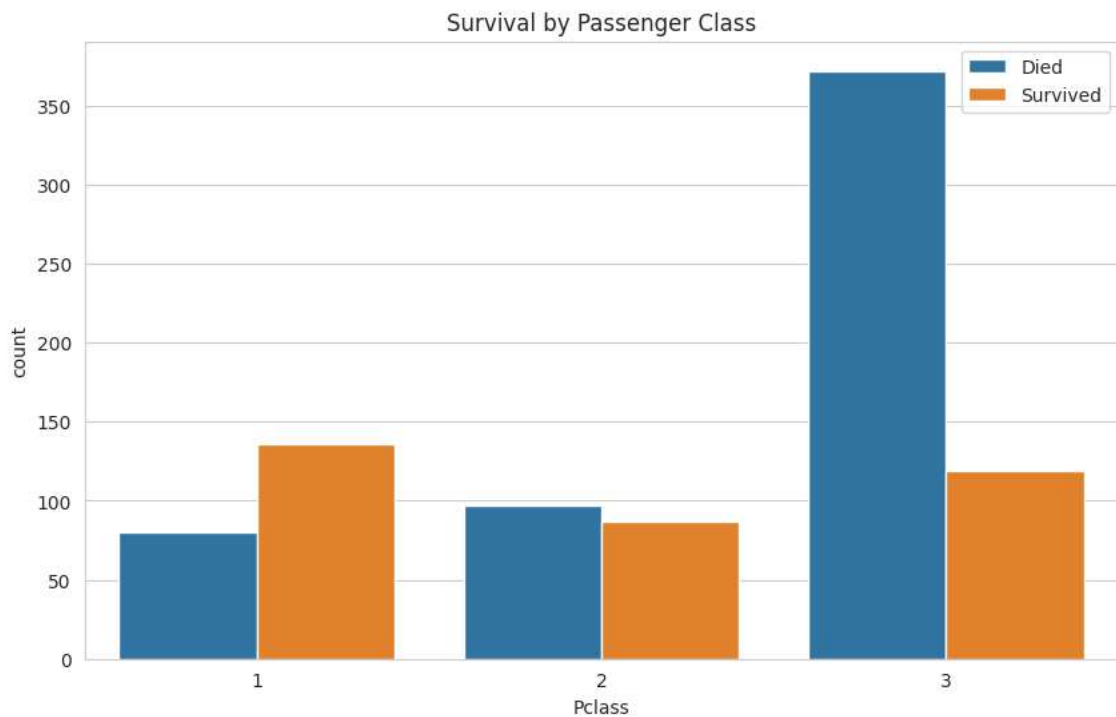
The average age of a passenger was 29.7 years.

The overall survival rate was 38.4%.

Fares ranged from \$0.0 to a whopping \$512.3292.

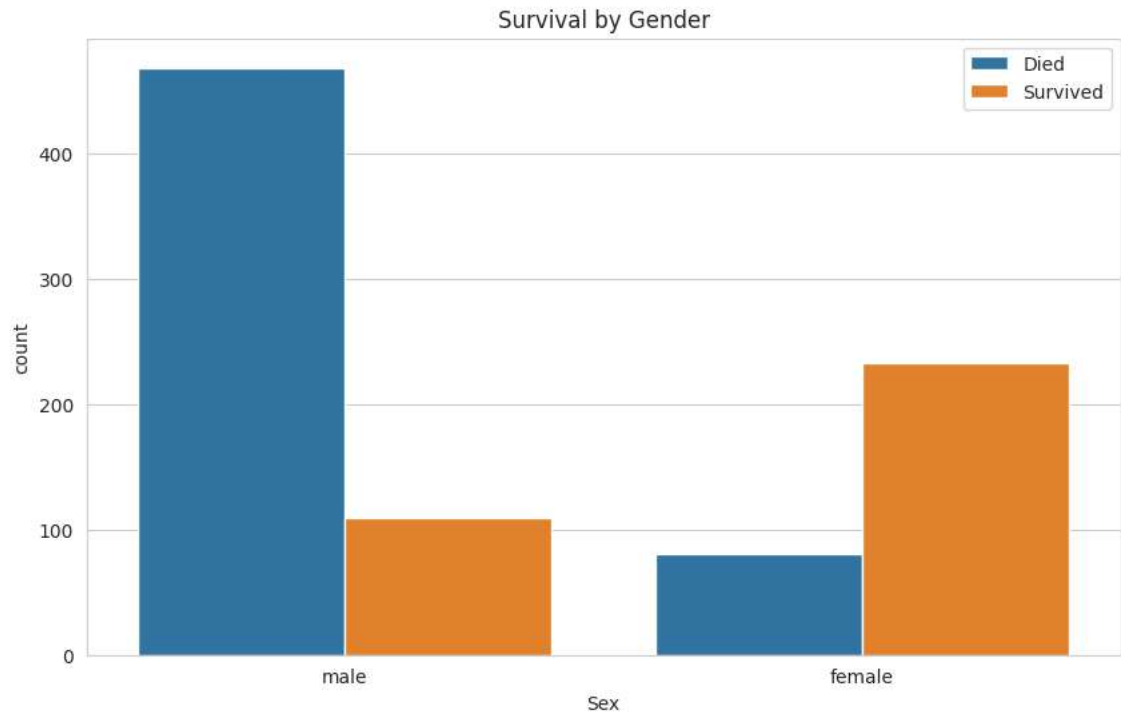


Insight: Far more people died than survived. This is an example of an imbalanced dataset, which can sometimes be a challenge for machine learning models.

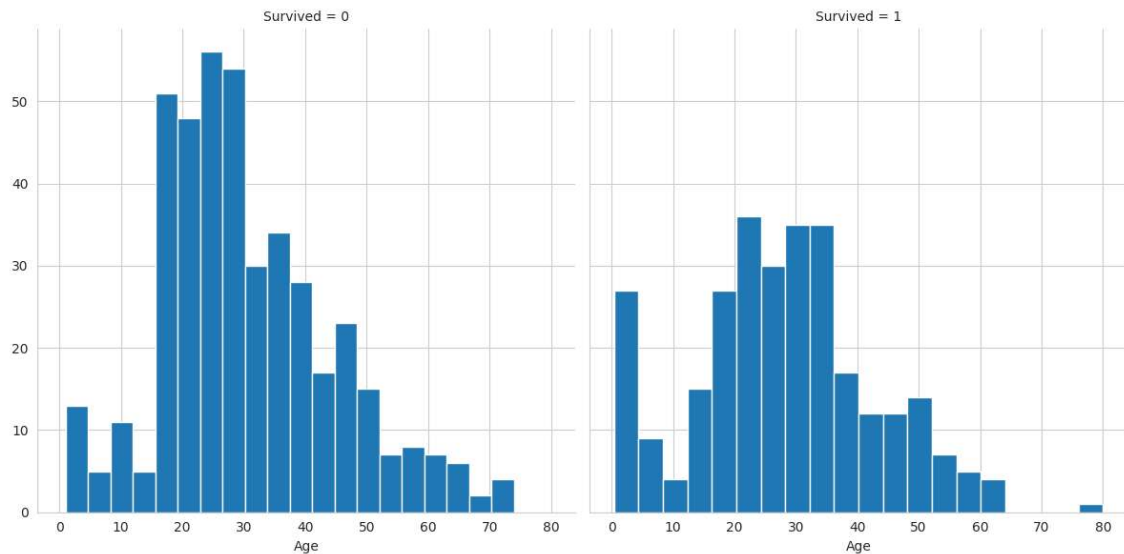


Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compared to 3rd class passengers. Money seems to

have made a difference.

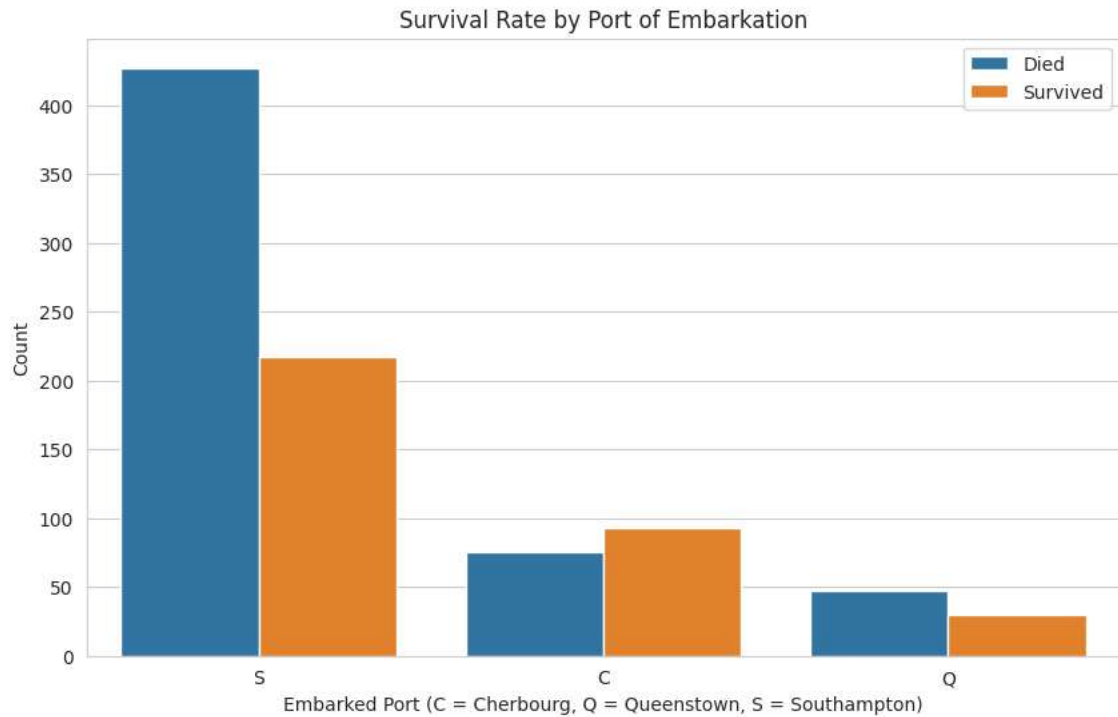


Insight: The pattern is undeniable. A much higher proportion of females survived compared to males. This is another very strong predictor.

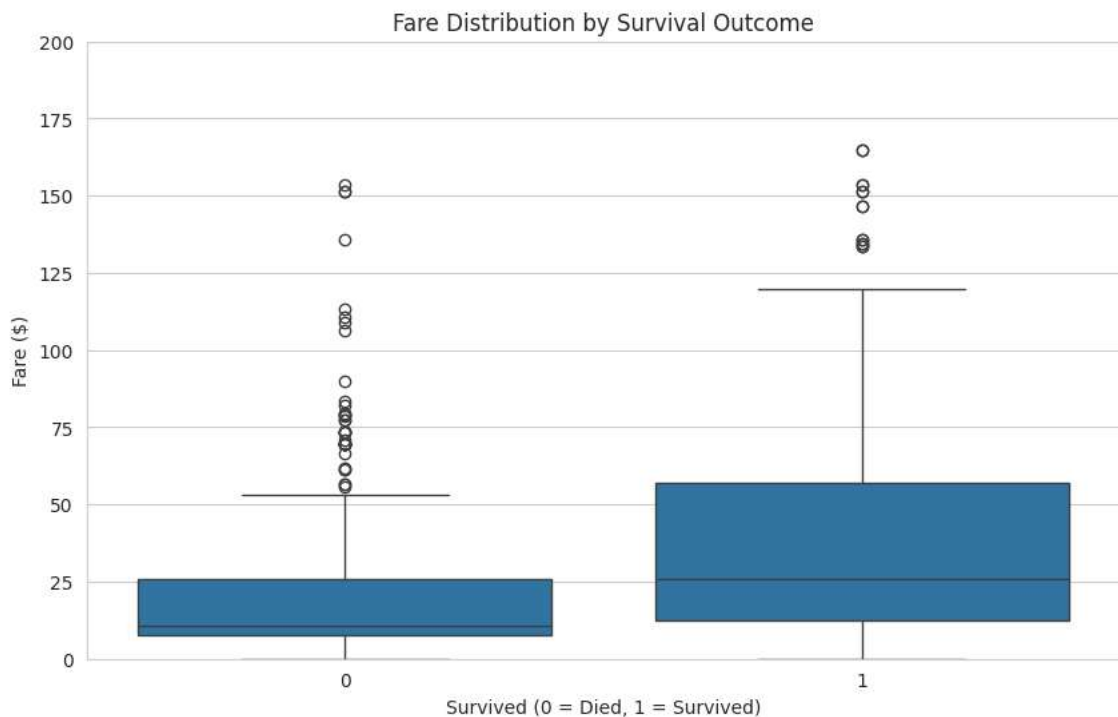


Insight: The age distribution for those who did not survive is centered around the 20-40 age range. For those who survived, there is a noticeable spike for young children. This supports the 'children' part of the mantra.

Experiment 1: Port of Embarkation



Experiment 2: Fare vs Survival



Observations & Findings:

Experiment 1: Passengers who boarded at Cherbourg (C) had a higher proportion of survival compared to those who boarded at Southampton (S) or Queenstown (Q). This is likely because a larger proportion of Cherbourg passengers booked 1st Class tickets.

Experiment 2: The boxplot demonstrates that passengers who survived paid a significantly higher median fare than those who died. The distribution for survivors extends to much higher

fare values, confirming that higher-paying passengers had better access to lifeboats.

Knowledge Check

1. What is the primary goal of Exploratory Data Analysis (EDA): The primary goal of EDA is to understand the underlying structure, patterns, anomalies, and relationships within a dataset before applying formal modeling or machine learning algorithms. It helps identify key feature interactions, detect missing data/outliers, and validate domain assumptions.

2. Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic: Based on the visual analysis:

Class: 1st Class passenger

Gender: Female

Age: Young child (under 10 years old)

Thus, a young female in 1st Class had the highest statistical probability of survival.

3. Why is it important to visualize data instead of just looking at summary statistics? What can a plot show you that a number like 'mean' or 'count' can't?

Visualizing data rather than relying solely on summary statistics is essential because single metrics like mean or count can obscure critical underlying patterns in a dataset. Summary statistics collapse complex data into single values, which conceals skewed distributions, bimodal patterns such as the high survival spike among young children that a mean age of 29.7 years hides, and extreme outliers, like the \$512.33 maximum fare that heavily inflates the average fare statistic. Plots overcome these limitations by revealing the complete shape of distributions, identifying outliers via visual ranges, and uncovering nuanced interactions between multiple variables, such as how survival rates differ by gender across different passenger classes.